

Muhamad Assem Ahmed

Lab 17

Part A

LAB 17 — Lab ID: 27 | MIN_GPA_EDIT = 2.6 | MAX_YEAR_EDIT = 2

Create()'s GET overload isn't async because it does nothing but return an empty View() no database call and no await so there's no asynchronous work to wait on The POST overload calls AddAsync SaveChangesAsync which is why it needs async

Part B

← → ↻ 🔒 <https://localhost:7273/students/edit/1> ☆ ... 🔍 Search

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Edit a Student

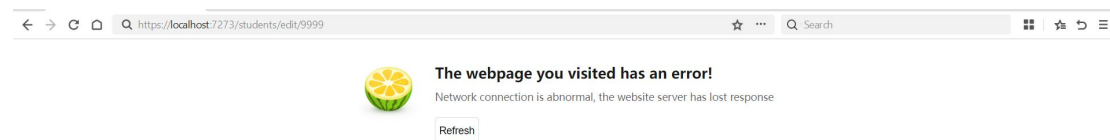
Everything typed here is sent to the POST version of **Create**.

Full Name

Year of Study

GPA

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This loads the student from the database instead of rendering an empty form because Edit's whole point is to show what already exists

Part C

Before Update

← → ↻ 🔒 <https://localhost:7273> ☆ ... 🔍 Search

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Student Portal

Every row below travelled from SQL Server, through Entity Framework, into a controller that never called **new**.

Today: the same rows, reachable at </students> — an address we design ourselves.

Full Name	Year	GPA
Kareem Fouad	4	3.20
Nada Samir	1	3.60
Omar Hesham	3	2.80
Yara Adel	2	3.50

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After :

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Yara Adel updated

All Students

Reached through the route pattern `students`.

+ Add a Student

Id	Full Name	Year	GPA
4	Kareem Fouad	4	3.20
3	Nada Samir	1	3.60
2	Omar Hesham	3	2.80
1	Yara Adel	2	3.10

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If id were renamed to `studentId` while the route still says `{id}`, model binding (which matches by NAME, not position) would find nothing called `studentId` in the route values and default it to 0. Since `student.Id` came from the form and is the real id, the guard clause's `'id' != student.Id` check would now always be true every edit would silently get rejected

Part D

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Edit a Student

Everything typed here is sent to the POST version of `Edit`.

Full Name

Year of Study

GPA

GPA must be at least 2.6 for an edit.

Save Cancel

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[Range] is validation-only, not a schema attribute: changing its bounds never produces a pending EF Core migration, because it's never mapped to a database

Part E

I edited name to be include 'edit for part E'

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Yara Adel edit for part E

Reached through the route pattern `students/{id:int}`.

Id	1
Full Name	Yara Adel edit for part E
Year of Study	2
GPA	3.10

[Back to all students](#)

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part F

Lab ID 27. $\text{MIN_GPA_EDIT} = 2.0 + (27 \bmod 5) \times 0.3 = 2.0 + 0.6 = 2.6$

$\text{MAX_YEAR_EDIT} = (27 \bmod 3) + 2 = 0 + 2 = 2$

Route constraint(before any code runs) checks the shape of the URL.

Validation / ModelState (before the action's first line) checks the shape of the data. Part D

Database constraint (at save) checks the truth of the data.

Before the save, the address bar held a POST to /Students/Edit, so F5 re-sent that POST. After redirecting to MuhamadAssemConfirmed the address bar now holds a GET to that action's address

both [Required]/[MaxLength] and [Range] are written as plain C#

attributes directly above a property. Different: [Required]/[MaxLength] are schema attributes — EF Core maps them onto the actual database column both

[Required]/[MaxLength] and [Range] are written as plain C#

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while [Range] is validation-only never touches the database schema while [Range] is validation-only and never touches the database schema